MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

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Abstract. Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X, with modular characteristic $\kappa = \kappa(\mathcal{A})$, cubic shadow α, and quartic shadow S_4. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ defines a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra, whose finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$ form the *shadow Postnikov tower*.

We prove four results. First, on each one-dimensional primary line L in the cyclic deformation complex, the weighted shadow generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies the algebraic relation $H(t)^2 = t^4 Q(t)$, where $Q(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ is the *shadow metric*: the tower is algebraic of degree 2 along L, and every coefficient S_r for $r \geq 5$ is determined by the three seed invariants (κ, α, S_4) of L. Second, the *critical discriminant* $\Delta = 8\kappa S_4$ partitions the standard Lie-theoretic landscape into four depth classes: Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$), with $\Delta = 0$ on the primary line forcing finite depth and $\Delta \neq 0$ forcing infinite depth. Third, we compute the genus-2 free energy $F_2 = \kappa \cdot 7/5760$ for single-generator algebras by an explicit stable-graph sum, carry out the first multi-channel computation for $\mathcal{W}_3$ (exhibiting a cross-channel correction whose cancellation is equivalent to multi-generator universality at genus 2), and derive the planted-forest correction $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$. Fourth, we prove that the gravitational coproduct, obtained by homological perturbation transfer through BRST reduction from $\widehat{\mathfrak{sl}}_2$ to Virasoro, is strictly primitive at all arities: gravity does not produce particles, only braids. The mechanism (ghost-number obstruction) applies to all principal Drinfeld–Sokolov reductions.

1. Introduction

1.1. **The question.** Classical Koszul duality is a phenomenon of graded algebras over a point. A quadratic algebra A determines a dual coalgebra $A^!$; the bar construction $B(A)$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence

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of a linear resolution. The theory is the subject of Priddy [15], Beilinson–Ginzburg–Soergel [2], and Loday–Vallette [13].

On an algebraic curve X, the generators of the algebra become sections of a $\mathcal{D}_X$-module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic 1-form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space. The resulting object, the *chiral bar complex* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$, is a factorization coalgebra on the Ran space $\mathrm{Ran}(X)$ [1, 6, 4].

At genus 0, the classical theory embeds faithfully. At genus $g \geq 1$, the topology of the curve forces genuinely new structure: the bar differential acquires curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and the scalar $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

1.2. **The answer: the shadow tower.** The bar complex on a curve of genus g with n marked points defines a pronilpotent dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$ on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, and the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is therefore a Maurer–Cartan element: $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

The MC element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. Its finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$, obtained by truncating the arity filtration at level r, form the *shadow Postnikov tower*. At arity 2, the tower produces the modular characteristic κ. At arity 3, a cubic shadow α (the structure constant of the transferred L_∞ bracket ℓ_3). At arity 4, a quartic shadow S_4 (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating $\Theta_{\mathcal{A}}$ at finite arity is measured by classes $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\mathrm{sh}})$ in the shadow algebra $\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A}))$. When $o_{r+1} = 0$, the tower stops; when $o_{r+1} \neq 0$, the next shadow coefficient S_{r+1} is forced. The question is how far this process continues and what controls it.

1.3. **The shadow metric and algebraicity.** The first main result (Theorem 4.1, parts (i)–(ii)) is that the shadow tower is controlled by a single quadratic polynomial.

Restrict to a one-dimensional primary slice L of the cyclic deformation complex (the space spanned by a single primary field η of definite conformal weight). On L, the MC recursion reduces to a scalar recursion for the shadow coefficients S_r, and the weighted generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$

satisfies the algebraic relation

$$H(t)^2 \;=\; t^4\, Q(t), \tag{1.1}$$

where the *shadow metric*

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2 \tag{1.2}$$

is a quadratic polynomial whose coefficients are determined by the three seed invariants (κ, α, S_4) of the algebra. The generating function is therefore algebraic of degree 2: $H(t) = t^2\sqrt{Q(t)}$. Every shadow coefficient S_r for $r \geq 5$ is a rational function of (κ, α, S_4), computable from the binomial expansion of $\sqrt{Q}$.

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of $F(t) = H(t)/t^2$, showing $F^2 = Q$. The orders $m = 0, 1, 2$ of this identity reproduce the definitions of κ, α, S_4; the orders $m \geq 3$ are exactly the recursion.

This is the *intrinsic quartic principle*: an infinite sequence of invariants, one for each arity $r \geq 2$, is determined by three numbers. The situation is analogous to the role of the Weierstrass equation for an elliptic curve: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here, three parameters (κ, α, S_4) determine all shadow coefficients S_r, because H satisfies an algebraic equation of degree 2 in H.

1.4. **The four-class classification.** Write $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$ for the *Gaussian decomposition*, where $\Delta := 8\kappa S_4$ is the *critical discriminant*. The Gaussian envelope $(2\kappa + 3\alpha t)^2$ is the part of Q visible to the quadratic and cubic shadows alone; Δ measures the irreducible quartic contribution.

The critical discriminant controls a fundamental dichotomy.

When $\Delta = 0$, the shadow metric is a perfect square: $Q = (2\kappa + 3\alpha t)^2$, the generating function $H(t) = t^2(2\kappa + 3\alpha t)$ is polynomial, and the tower terminates. If $\alpha = 0$, only $S_2 = \kappa$ survives (depth 2); if $\alpha \neq 0$, then S_2 and S_3 survive (depth 3).

When $\Delta \neq 0$, the shadow metric is irreducible: $\sqrt{Q}$ is irrational, every Taylor coefficient of $\sqrt{Q}$ beyond degree 1 carries a factor of Δ, and $S_r \neq 0$ for all $r \geq 4$ (generically). The tower is infinite.

This yields the second main result (Theorem 4.1, part (iii)): every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes*. The partition is exhaustive for the standard Lie-theoretic landscape (Table 1); whether every chirally Koszul algebra belongs to one of these four classes is a structural observation, not a classification theorem.

Class	Δ	$r_{\max}$	Prototype	Shadow data
G (Gaussian)	0	2	Heisenberg	$\kappa = k$
L (Lie)	0	3	Affine Kac–Moody	κ, $\alpha \neq 0$
C (Contact)	0^*	4	$\beta\gamma$ system	$\kappa = 6\lambda^2 - 6\lambda + 1$
M (Mixed)	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$	κ, α, S_4

*Class **C**: $\Delta = 0$ on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G** consists of algebras whose bar complex is formal in the classical sense: the shadow tower terminates at the leading scalar, and all higher-genus free energies F_g are determined by κ alone. Class **L** adds exactly one cubic correction (the structure constant of the Lie bracket on generators), after which the tower terminates by the Jacobi identity. Class **C** is the first nonlinear class: the quartic contact invariant S_4 is nonzero but lives on a separate stratum whose self-bracket exits the complex, so the tower terminates after one further step. Class **M** is the generic class: the shadow tower is infinite, the free energies F_g receive corrections at every arity, and the shadow coefficients S_r decay (or grow) at a computable rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) \;=\; \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}. \tag{1.3}$$

The growth rate is the reciprocal of the branch-point modulus of the shadow metric. When $\rho < 1$, the shadow tower converges absolutely; when $\rho > 1$, it diverges but the algebraic function $\sqrt{Q}$ extends analytically past the convergence radius. For the Virasoro algebra at central charge c:

$$\rho(\mathrm{Vir}_c) \;=\; \frac{1}{c}\sqrt{\frac{180c + 872}{5c + 22}}, \tag{1.4}$$

with a critical central charge $c_\star \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separating the convergent and divergent regimes. At the self-dual point $c = 13$ (where $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ satisfies $\mathrm{Vir}_c = \mathrm{Vir}_c^!$): $\rho \approx 0.467$.

Classical Koszul duality over a point is the genus-0, arity-2, $\Delta = 0$ stratum of this picture: the formal Gaussian case where the tower terminates immediately. Everything beyond class **G** is new.

1.5. **The shadow connection and Koszul monodromy.** The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\mathrm{sh}} \;=\; d \;-\; \frac{Q'(t)}{2\,Q(t)}\,dt, \tag{1.5}$$

with regular singularities at the zeros of Q and residue $\frac{1}{2}$. The monodromy around each branch point is -1. This is the *Koszul sign*: the square root

$\sqrt{Q}$ changes sign around a simple zero, and the shadow tower inherits this sign change through the algebraic relation $H = t^2\sqrt{Q}$.

Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow metric transforms by the Verdier involution. For the Virasoro family $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$, the discriminants satisfy

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) \;=\; \frac{6960}{(5c+22)(152-5c)}, \tag{1.6}$$

a Virasoro-specific identity whose numerator $6960 = 40{\cdot}174$ is a product of the Virasoro quartic normalization and the sum of the Lee–Yang denominators. The Koszul dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. For other families (affine Kac–Moody, $\mathcal{W}_N$ with $N \geq 3$), the discriminant sum has a different numerator.

1.6. **Genus-2 free energies.** The shadow tower produces invariants of moduli spaces by evaluating graph sums over stable curves. The genus-g *free energy* $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is computed for each single-generator chirally Koszul algebra at all genera (and unconditionally at genus 1) by the *Faber–Pandharipande integral* $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\,\lambda_g$, where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle and ψ_1 is the cotangent line class.

At genus 2: $\lambda_2^{\mathrm{FP}} = 7/5760$, and $F_2(\mathcal{A}) = \kappa \cdot 7/5760$ for every single-generator chirally Koszul algebra. The verification requires an explicit sum over the six stable graphs of genus 2 with no marked points (the smooth genus-2 curve, figure-eight, banana, dumbbell, theta, and lollipop), weighted by automorphism factors and vertex contributions from the shadow tower.

For the $\mathcal{W}_3$ algebra, which has two generators T (weight 2) and W (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label (T or W), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator $P_T = 2/c$, $P_W = 3/c$ depends on the channel. The per-channel diagonal sum yields $F_2^{\mathrm{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\mathrm{FP}} = 7c/6912$, but the full multi-channel graph sum produces an additional cross-channel correction $\delta F_2 = (c+120)/(16c)$. Whether this correction is cancelled by CohFT R-matrix terms (restoring the universality identity $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at genus 2 for multi-generator algebras) is the content of the multi-generator universality problem, which remains open.

The Maurer–Cartan equation on $\overline{\mathcal{M}}_{2,0}$ also produces a *planted-forest correction*

$$\delta_{\mathrm{pf}}^{(2,0)} \;=\; \frac{S_3(10\,S_3 - \kappa)}{48}, \tag{1.7}$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where $S_3 = 0$). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra: $\delta_{\mathrm{pf}}^{(2,0)} = -(c-40)/48$.

1.7. **The gravitational coproduct.** The fourth result concerns the transferred algebraic structure under BRST reduction. The Drinfeld–Sokolov (DS) reduction functor takes affine Kac–Moody data to $\mathcal{W}$-algebra data via a BRST complex with ghosts. The homological perturbation lemma (HPL) transfers the A_∞ products and the coproduct from the affine dg-shifted Yangian to a transferred Yangian on the $\mathcal{W}$-algebra sector.

We prove the result for the principal DS reduction $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$. The ghost-number obstruction (the homotopy h in the SDR has ghost degree -1, and the projection $p \otimes p$ kills all ghost-number-(-1) elements) forces the transferred coproduct to be *strictly primitive at all arities*. The mechanism applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$, since the ghost-number grading of the BRST complex is universal; we state the general case as a conjecture and prove it for $\widehat{\mathfrak{sl}}_2$:

$$\Delta_z^{\mathrm{Vir}}(T) \;=\; \tau_z(T) \otimes 1 + 1 \otimes T, \qquad \Delta_{z,n}^{\mathrm{Vir}} = 0 \quad \text{for } n \geq 2.$$

All higher-arity coproduct corrections, which would represent graviton splitting into constituent gravitons, are killed by the ghost-number mechanism. The only nontrivial gravitational structure is the r-matrix twist $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ on the target product, encoding the braiding of line operators.

The physical content: in Chern–Simons gauge theory, gluons split (the coproduct is nontrivial, reflecting the Clebsch–Gordan decomposition of tensor products). In 3d gravity, gravitons do not split. The gravitational coproduct is primitive; all interactions are encoded in the twist, not in particle production.

1.8. **Relation to existing work.** The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [1] and developed in the factorization algebra framework by Francis and Gaitsgory [6] and Costello and Gwilliam [4]. The modular characteristic κ appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula $\kappa(\mathrm{Vir}_c) = c/2$ can be extracted from Beilinson and Schechtman [3]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [7], Mumford [14], and the extensive literature on cohomological field theories following Kontsevich and Manin [10].

The quartic shadow S_4, the critical discriminant Δ, the four-class partition, and the algebraicity theorem (Theorem 4.1) appear to be new. The connection between shadow depth and L_∞ non-formality, while consonant with the general philosophy of Kontsevich [11], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [12, 8, 5]; the gravitational coproduct primitivity result (Theorem 8.2) and its connection to the shadow tower appear to be new.

1.9. **Structure of the paper.** Section 2 reviews the shadow Postnikov tower, primary lines, and the shadow metric, establishing notation and the three seed invariants (κ, α, S_4). Section 3 proves the Riccati algebraicity theorem, derives the closed-form generating function, the shadow connection, and the single-line dichotomy. Section 4 collects the main results as a single theorem. Section 5 proves the four-class classification and the universal factorization theorem. Section 6 computes (κ, α, S_4) for every standard family, and presents explicit shadow towers, complementarity data, and genus expansion tables. Section 7 carries out the genus-2 graph sum, including the multi-channel computation for $\mathcal{W}_3$ and the planted-forest correction. Section 8 proves the gravitational coproduct primitivity. Section 9 discusses open problems, including multi-generator universality at genus ≥ 3, the operadic complexity conjecture, the connection to Pixton's ideal, and the arithmetic content of shadow denominators.

2. The shadow metric and the Riccati equation

2.1. **The shadow Postnikov tower.** Let $\mathcal{A}$ be a modular Koszul chiral algebra with modular cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ and universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$. The complex carries an arity filtration $F^\bullet$ with $F^r = \bigoplus_{s \geq r} \mathrm{Def}^{(s)}_{\mathrm{cyc}}$, and the *shadow Postnikov tower* is the sequence of finite-order truncations

$$\Theta^{\leq 2}_{\mathcal{A}},\ \Theta^{\leq 3}_{\mathcal{A}},\ \Theta^{\leq 4}_{\mathcal{A}},\ \dots$$

where $\Theta^{\leq r}_{\mathcal{A}}$ solves the Maurer–Cartan equation in the quotient $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}/F^{r+1}$. The arity-$r$ projection $\mathrm{Sh}_r(\mathcal{A})$ is the *r-th shadow* of $\mathcal{A}$. On a one-dimensional slice of the deformation complex, Sh_r is determined by a single scalar S_r; the MC equation projected to arity r gives the *all-arity master equation*

$$\nabla_H(\mathrm{Sh}_r) + \mathfrak{o}^{(r)} = 0, \tag{2.1}$$

where ∇_H is the linearized operator and $\mathfrak{o}^{(r)}$ is the obstruction from lower arities.

2.2. **Primary lines and the shadow metric.**

Definition 2.1 (Primary line)**.** A *primary line* in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ is a one-dimensional subspace $L = \mathbf{k} \cdot \eta$ spanned by a single primary field η of definite conformal weight h_η. For a chiral algebra with generators $\phi_1, \dots, \phi_N$ of weights $h_1, \dots, h_N$, each generator spans a primary line; the full first-order deformation space is $\bigoplus_i L_i$.

Let L be a primary line with coordinate x, and write $\mathrm{Sh}_r|_L = S_r\, x^r$ for the shadow data restricted to L. Set

$$\kappa := S_2, \qquad \alpha := S_3.$$

The scalar κ is the modular characteristic (curvature) of $\mathcal{A}$ on L; the scalar α is the cubic shadow.

Definition 2.2 (Shadow metric)**.** Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* (shadow quadratic form) on L is the polynomial

$$Q_L(t) \;:=\; a_0^2+2\,a_0\,a_1\,t+(a_1^2+2\,a_0\,a_2)\,t^2 \;=\; 4\kappa^2+12\kappa\alpha\,t+(9\alpha^2+16\kappa S_4)\,t^2 \;\in\; k(c)[t]. \tag{2.2}$$

The *critical discriminant* of L is

$$\Delta := 8\kappa S_4 = a_0\,a_2. \tag{2.3}$$

Remark 2.3. The term "metric" is used by analogy with the role of Q_L in governing the geometry of the shadow tower, not in the sense of a Riemannian metric. The shadow metric is a quadratic polynomial in the arity parameter t, determined by three OPE invariants: κ, α, and S_4.

2.3. **The recursion on a primary line.** On a primary line L, the all-arity master equation (2.1) reduces to a recursion among the scalars S_r.

Proposition 2.4 (Single-line recursion)**.** *On a primary line L with propagator $P := 1/\kappa$, the MC equation at arity $r \geq 5$ is*

$$S_r \;=\; -\frac{P}{2r}\sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\,j\,k\,S_j\,S_k, \qquad c_{jk} = \begin{cases} 1 & \text{if } j<k,\\ 1/2 & \text{if } j=k. \end{cases} \tag{2.4}$$

This recursion is well-defined: since $j \geq 3$ and $j+k = r+2$, every term on the right has $k = r+2-j \leq r-1$, so S_r is determined by $\{S_2, S_3, S_4, \ldots, S_{r-1}\}$. The arity-4 shadow S_4 is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside $\kappa = S_2$ and $\alpha = S_3$.

Proof. The Lie bracket on the cyclic deformation complex, restricted to L, pairs $\mathrm{Sh}_j = S_j x^j$ and $\mathrm{Sh}_k = S_k x^k$ to produce a term of arity $j+k-2$ with coefficient $jk\,P\,S_jS_k$. The eigenvalue identity $\nabla_H(x^r) = 2r\,x^r$ separates the MC equation at arity r into the linear term $2r\,S_r$ on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor c_{jk} accounts for the factor $\frac{1}{2}$ in $\frac{1}{2}[\Theta,\Theta]$ when $j=k$. □

2.4. **The weighted shadow generating function.**

Definition 2.5 (Weighted shadow generating function)**.** The *weighted shadow generating function* on L is

$$H(t) \;:=\; \sum_{r\geq 2} r\,S_r\,t^r. \tag{2.5}$$

Setting $a_n := (n+2)\,S_{n+2}$ for $n \geq 0$, this becomes $H(t) = \sum_{n\geq 0} a_n\,t^{n+2}$, with initial values $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$.

The weighted generating function H packages the recursion (2.4) as a convolution identity. Define $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\,t^n$. The Cauchy product $F^2 = \sum_{m\geq 0}(\sum_{i+j=m} a_ia_j)\,t^m$ has its coefficients at $m \geq 3$ controlled

by the MC recursion. The next theorem shows that the MC equation forces F^2 to be a polynomial of degree 2.

3. Algebraicity and the depth classification

3.1. The Riccati algebraicity theorem.

Theorem 3.1 (Riccati algebraicity)**.** *Let L be a primary line with shadow data S_r, shadow metric Q_L (Definition 2.2), and weighted shadow generating function H (Definition 2.5). Then the all-arity master equation on L is equivalent to the algebraic relation*

$$H(t)^2 \;=\; t^4\, Q_L(t). \tag{3.1}$$

The shadow generating function is therefore algebraic of degree 2 over $k(\kappa,\alpha,S_4)(t)$:

$$H(t) \;=\; t^2\sqrt{Q_L(t)}. \tag{3.2}$$

Proof. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\, t^n$ where $a_n = (n+2)\, S_{n+2}$. The claim (3.1) is equivalent to $F(t)^2 = Q_L(t)$. We verify this by computing the Cauchy product $F^2 = \sum_{m\geq 0}\big(\sum_{i+j=m} a_i a_j\big)\, t^m$ and showing it agrees with Q_L at all orders.

Orders 0, 1, 2 *(initial data).* At $m=0$: $a_0^2 = (2\kappa)^2 = 4\kappa^2$. At $m=1$: $2\, a_0\, a_1 = 2\cdot 2\kappa\cdot 3\alpha = 12\kappa\alpha$. At $m=2$: $a_1^2 + 2\, a_0\, a_2 = 9\alpha^2 + 2\cdot 2\kappa\cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$. These are exactly the coefficients of $Q_L(t)$ in (2.2).

Orders $m\geq 3$ (the MC recursion forces vanishing). Since Q_L has degree 2 in t, the identity $F^2 = Q_L$ requires

$$\sum_{i+j=m} a_i\, a_j \;=\; 0 \qquad \text{for all } m\geq 3. \tag{3.3}$$

We show that this is exactly the MC recursion (2.4) at arity $r = m+2$.

The sum $\sum_{i+j=m} a_i a_j$ with $a_i = (i+2)\, S_{i+2}$ and $a_j = (j+2)\, S_{j+2}$ equals $\sum_{p+q=m+4,\, p,q\geq 2} p\, q\, S_p\, S_q$, where $p = i+2$, $q = j+2$. Setting $r = m+2$, this becomes $\sum_{p+q=r+2,\, p,q\geq 2} p\, q\, S_p\, S_q$.

Isolating the terms with $p=2$ or $q=2$ (which contribute 2κ from $S_2=\kappa$):

$$\begin{aligned}\sum_{p+q=r+2,\, p,q\geq 2} pq\, S_pS_q &= 2\cdot 2r\cdot\kappa\, S_r + \sum_{\substack{p+q=r+2\\ p,q\geq 3}} pq\, S_pS_q\\ &= 4r\kappa\, S_r + 2\sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_jS_k,\end{aligned}$$

where the factor 2 in front of the restricted sum accounts for $S_pS_q + S_qS_p$ when $p\neq q$, and c_{jk} corrects for the diagonal. Substituting the recursion (2.4):

$$2\sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_jS_k \;=\; 2\cdot(-2r\kappa\, S_r) \;=\; -4r\kappa\, S_r.$$

Hence $\sum_{p+q=r+2} pq\, S_pS_q = 4r\kappa S_r - 4r\kappa S_r = 0$, which is (3.3).

Conversely, the vanishing (3.3) at order $m \geq 3$ implies the recursion (2.4) at $r = m + 2$ by solving for S_r (using $\kappa \neq 0$).

The equivalence of (3.1) with the full MC recursion is therefore established at all arities. Taking positive square roots (the branch with $F(0) = a_0 = 2\kappa > 0$) gives $H(t) = t^2 F(t) = t^2 \sqrt{Q_L(t)}$. □

Remark 3.2 (Riccati ODE reformulation). Set $U(t) := \sum_{r\geq 3} S_r\, t^r$ (the non-linear part, removing Sh_2). The algebraic relation (3.1) is equivalent to the Riccati-type first-order ODE

$$2t\, U'(t) + \frac{P}{2}\big(U'(t)\big)^2 = R_3\, t^3 + R_4\, t^4, \tag{3.4}$$

where $R_3 = 6\alpha$ and $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$ encode the cubic and quartic inputs. The left-hand side is quadratic in U': this is the precise sense in which the shadow tower is algebraic of degree 2.

3.2. **Gaussian decomposition.**

Corollary 3.3 (Gaussian decomposition)**.** *The shadow metric decomposes as*

$$Q_L(t) \;=\; (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2, \tag{3.5}$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. The first term $(2\kappa + 3\alpha t)^2$ is the Gaussian envelope*: the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term $2\Delta\, t^2$ is the* interaction correction*, measuring the quartic OPE contribution not captured by the cubic data alone.*

Proof. $(2\kappa+3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha\, t + 9\alpha^2\, t^2$. Adding $2\Delta\, t^2 = 16\kappa S_4\, t^2$ recovers the coefficient $9\alpha^2 + 16\kappa S_4$ of t^2 in $Q_L(t)$. □

3.3. **Intrinsic quartic principle.**

Corollary 3.4 (Intrinsic quartic principle)**.** *On a primary line L, the shadow tower has intrinsic OPE data at arities 2, 3, 4 only. All shadow coefficients S_r for $r \geq 5$ are determined by (κ, α, S_4) alone, via the closed form*

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\,\sqrt{Q_L(t)} \qquad (r \geq 2). \tag{3.6}$$

Proof. $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ depends on exactly (κ, α, S_4). Since $H(t) = t^2\sqrt{Q_L(t)} = \sum_{r\geq 2} r S_r\, t^r$ by Theorem 3.1, comparing coefficients gives $rS_r = [t^{r-2}]\sqrt{Q_L}$. □

3.4. **The shadow connection.**

Proposition 3.5 (Shadow connection)**.** *The shadow metric defines a logarithmic connection on $\mathcal{O}_L$ over $L \setminus \{Q_L = 0\}$:*

$$\nabla^{\mathrm{sh}} \;=\; d - \omega, \qquad \omega \;:=\; \frac{Q_L'(t)}{2\, Q_L(t)}\, dt \;=\; d\big(\log\sqrt{Q_L}\big). \tag{3.7}$$

(i) *The connection is flat:* $(\nabla^{\mathrm{sh}})^2 = 0$.

(ii) *At each simple zero t_0 of Q_L, the residue of ω is $\frac{1}{2}$.*
(iii) *The monodromy is $\exp(2\pi i \cdot \frac{1}{2}) = -1$: the standard monodromy of a square root around a simple branch point.*
(iv) *The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)}/(2\kappa)$, and the shadow generating function is $H(t) = 2\kappa\, t^2\, \Phi(t)$: the parallel transport of the curvature along the arity parameter, weighted by t^2.*

Proof. Flatness is automatic: ω is a closed 1-form on a 1-dimensional base. At a simple zero t_0 of Q_L: $Q_L(t) \approx Q_L'(t_0)(t - t_0)$, so $\omega \approx \frac{1}{2(t-t_0)}\, dt$, giving residue $\frac{1}{2}$. The monodromy exponentiation is immediate. For (iv): $\nabla^{\mathrm{sh}}\Phi = 0$ since $d\Phi = \omega\, \Phi$ by construction, and the evaluation at $t = 0$ gives $\Phi(0) = \sqrt{Q_L(0)}/(2\kappa) = 2\kappa/(2\kappa) = 1$. □

Remark 3.6. The shadow generating function $H(t) = 2\kappa\, t^2\, \Phi(t)$ is *not* a flat section of ∇^{sh}: the factor t^2 contributes $\nabla^{\mathrm{sh}}(H) \neq 0$. The flat section is $\Phi(t) = \sqrt{Q_L(t)}/(2\kappa)$; the shadow tower is the Taylor expansion of $t^2 \cdot \Phi(t)$, reflecting the arity offset (the tower starts at arity 2, not arity 0).

3.5. **The single-line dichotomy.**

Theorem 3.7 (Single-line dichotomy)**.** *Let L, S_r, α, Δ, Q_L be as above, and define the shadow depth $r_{\max}|_L := \max\{r : S_r \neq 0\}$ (with $r_{\max} = \infty$ if $S_r \neq 0$ for infinitely many r). Then $r_{\max}|_L \in \{2, 3, \infty\}$, classified by the shadow metric:*

(i) ***Class* G** *($\Delta = 0$, $\alpha = 0$). $Q_L = (2\kappa)^2$ is constant. $H(t) = 2\kappa\, t^2$. Only S_2 survives: $r_{\max} = 2$.*
(ii) ***Class* L** *($\Delta = 0$, $\alpha \neq 0$). $Q_L = (2\kappa + 3\alpha\, t)^2$ is a perfect square. $H(t) = t^2(2\kappa + 3\alpha\, t)$. S_2 and S_3 survive: $r_{\max} = 3$.*
(iii) ***Class* M** *($\Delta \neq 0$). Q_L is irreducible over $k(c)$; $\sqrt{Q_L}$ is irrational; $S_r \neq 0$ for all $r \geq 4$ generically: $r_{\max} = \infty$.*

Moreover, for $r \geq 4$,

$$S_r = \Delta \cdot R_r, \qquad R_r \in \mathbb{Q}(\alpha, \kappa, S_4). \tag{3.8}$$

Proof. The classification follows from Theorem 3.1 and the Gaussian decomposition (Corollary 3.3).

Case $\Delta = 0$. Then $Q_L = (2\kappa+3\alpha t)^2$ is a perfect square, so $\sqrt{Q_L} = 2\kappa+3\alpha t$ is polynomial and $H(t) = t^2(2\kappa + 3\alpha t)$ terminates. If $\alpha = 0$: $H(t) = 2\kappa t^2$, giving $r_{\max} = 2$. If $\alpha \neq 0$: $H(t) = 2\kappa t^2 + 3\alpha t^3$, giving $r_{\max} = 3$.

Case $\Delta \neq 0$. The discriminant of Q_L as a polynomial in t is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so Q_L is irreducible over $k(c)$. The Taylor expansion of $\sqrt{Q_L}$ around $t = 0$ has nonzero coefficients at all orders generically, hence $r_{\max} = \infty$.

Universal factorization. The binomial expansion of $\sqrt{Q_L}$ around the Gaussian envelope $(2\kappa + 3\alpha t)$ is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t)\sqrt{1 + \frac{2\Delta\, t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding $\sqrt{1+u}$ as $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \cdots$ with $u = 2\Delta t^2/(2\kappa + 3\alpha t)^2$, every coefficient of t^n for $n \geq 2$ in $\sqrt{Q_L} - (2\kappa + 3\alpha t)$ carries a factor of Δ. Since $rS_r = [t^{r-2}]\sqrt{Q_L}$ for $r \geq 2$, and the Gaussian part $2\kappa + 3\alpha t$ contributes only at $r \leq 3$, the factorization $S_r = \Delta \cdot R_r$ for $r \geq 4$ follows. □

Remark 3.8 (The contact class). The partition **G**/**L**/**C**/**M** of modular Koszul algebras includes a fourth class **C** (contact, $r_{\max} = 4$), realized by the $\beta\gamma$ system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}$ where $\kappa = 0$, and the shadow metric requires $\kappa \neq 0$. On the charged stratum, the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at $r = 4$. The algebraic shadow depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ exhausts the four classes: **G** ($d_{\mathrm{alg}} = 0$), **L** ($d_{\mathrm{alg}} = 1$), **C** ($d_{\mathrm{alg}} = 2$), **M** ($d_{\mathrm{alg}} = \infty$).

3.6. **The shadow growth rate.** For class **M** algebras, the shadow tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients S_r.

Definition 3.9 (Shadow growth rate)**.** Let L be a primary line of class **M** ($\Delta \neq 0$), and write $Q_L(t) = q_0 + q_1 t + q_2 t^2$ for the shadow metric. The *shadow growth rate* of L is

$$\rho_L \;:=\; \sqrt{\frac{q_2}{q_0}} \;=\; \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}. \tag{3.9}$$

By Vieta's theorem, the zeros t_0, $\bar{t}_0$ of Q_L satisfy $|t_0|^2 = q_0/q_2$, so $\rho_L = 1/|t_0|$: the reciprocal of the branch-point modulus. For classes **G** and **L**, set $\rho := 0$ by convention.

Theorem 3.10 (Shadow asymptotics)**.** *Let L be a primary line of class* **M** *with shadow growth rate ρ_L.*

(i) Asymptotic formula.

$$S_r \;\sim\; A(\mathcal{A})\, \rho_L^r\, r^{-5/2} \cos(r\theta + \varphi) \qquad (r \to \infty), \tag{3.10}$$

where $\theta := -\arg(t_0)$ is the oscillation phase, $A(\mathcal{A}) > 0$, and φ is computable from the branch-point residue of $\sqrt{Q_L}$. The exponent $-5/2$ arises from $S_r = [t^{r-2}]\sqrt{Q_L}/r$ combined with the $n^{-3/2}$ transfer law for square-root singularities.

(ii) Unit circle criterion. *$\rho_L = 1$ if and only if $q_0 = q_2$, i.e.,*

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta. \tag{3.11}$$

Equivalently, the branch points of Q_L lie on the unit circle $|t| = 1$.

(iii) Convergence partition. $\rho_L < 1$*: the shadow tower converges absolutely.* $\rho_L > 1$*: the shadow tower diverges, but the algebraic function* $\sqrt{Q_L(t)}$ *extends analytically past the convergence radius.* $\rho_L = 1$*: marginal case (the critical surface).*

Proof. The function $\sqrt{Q_L(t)}$ has algebraic branch-cut singularities of order $\frac{1}{2}$ at each simple zero t_0 of Q_L. Since Q_L is quadratic with nonzero discriminant ($\Delta \neq 0$ implies $\mathrm{disc}(Q_L) = -32\kappa^2\Delta \neq 0$), the two zeros $t_0, \bar{t}_0$ are distinct with $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$.

The transfer theorem for algebraic singularities (Flajolet–Sedgewick, Theorem VI.1) gives $[t^n]\sqrt{Q_L} \sim C_0\, t_0^{-n}\, n^{-3/2}$ for each dominant singularity t_0 at distance $R = 1/\rho_L$ from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from $2\,\mathrm{Re}(C_0\, t_0^{-(r-2)})$. Since $S_r = [t^{r-2}]\sqrt{Q_L}/r$ (Corollary 3.4), the extra factor $1/r$ gives the stated $r^{-5/2}$ exponent.

For (ii): $\rho_L = 1$ iff $q_2/q_0 = 1$. Writing $q_0 = 4\kappa^2$ and $q_2 = 9\alpha^2 + 2\Delta$, the condition $q_0 = q_2$ factors as $4\kappa^2 - 9\alpha^2 = 2\Delta$. Part (iii) is immediate from the definition of the convergence radius $R = 1/\rho_L$. □

3.7. **The depth decomposition.**

Theorem 3.11 (Depth decomposition)**.** *The shadow depth of a modular Koszul algebra* $\mathcal{A}$ *decomposes as*

$$d(\mathcal{A}) \;=\; 1 + d_{\mathrm{arith}}(\mathcal{A}) + d_{\mathrm{alg}}(\mathcal{A}), \tag{3.12}$$

where:

(i) d_{arith} *counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function* $\widehat{Z}^c_{\mathcal{A}}$ *on* $\mathcal{M}_{1,1}$*;*

(ii) $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ *is the* algebraic depth*, determined by the single-line dichotomy* (*Theorem* 3.7) *and stratum separation* (*Remark* 3.8)*:*

$$d_{\mathrm{alg}} = \begin{cases} 0 & \textit{class } \mathbf{G}, \\ 1 & \textit{class } \mathbf{L}, \\ 2 & \textit{class } \mathbf{C}, \\ \infty & \textit{class } \mathbf{M}; \end{cases}$$

(iii) *depths* $d \geq 5$ *are purely arithmetic: they arise from* d_{arith}*, not from the* A_∞ *structure.*

Proof. The algebraic depth d_{alg} is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 3.7) gives $d_{\mathrm{alg}} \in \{0, 1, \infty\}$ on each primary line, and stratum separation (Remark 3.8) provides $d_{\mathrm{alg}} = 2$ for the contact class. Since $d_{\mathrm{alg}} \leq 2$ for finite cases ($\mathbf{G}/\mathbf{L}/\mathbf{C}$) and the base term is 1 (the curvature κ at arity 2), the inequality $d_{\mathrm{alg}} \leq 4$ forces all depths $d \geq 5$ to arise from d_{arith}.

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing $\widehat{Z}^c_{\mathcal{A}}$ via the Roelcke–Selberg spectral decomposition on $\mathcal{M}_{1,1}$. Each holomorphic eigenform of weight k contributes a critical line at $\mathrm{Re}(s) = k/2$; Maass cusp-form projections contribute no new critical lines. □

Example 3.12 (Standard families)**.** The shadow metric data for the standard families:

Family	κ	α	S_4	Δ	Class	ρ
Heisenberg $\mathcal{H}_k$	k	0	0	0	**G**	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{d(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	**L**	0
$\beta\gamma$ (weight λ)	$6\lambda^2-6\lambda+1$	0	$\neq 0$	0^*	**C**	$*$
Virasoro Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	**M**	$\frac{1}{c}\sqrt{\frac{180c+872}{5c+22}}$

* Stratum separation: $\Delta = 0$ on the primary line (where $S_4 = 0$); the quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on a charged stratum where $\kappa = 0$, and the single-line dichotomy does not apply.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\mathrm{Vir}}(t) = c^2 + 12c\,t + \frac{180c+872}{5c+22}\,t^2,$$

with Gaussian decomposition $Q_{\mathrm{Vir}} = (c+6t)^2 + \frac{80}{5c+22}\,t^2$. The critical central charge c^* where $\rho = 1$ is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately $c^* \approx 6.12$. For $c > c^*$ the shadow tower converges; for $c < c^*$ it diverges. At the self-dual point $c = 13$: $\rho(\mathrm{Vir}_{13}) \approx 0.467$.

Remark 3.13 (Spectral curve)*.* The algebraic relation $H^2 = t^4 Q_L(t)$ defines, for each primary line L, a spectral curve $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}^2_{t,H}$. Since Q_L is quadratic in t, Σ_L has arithmetic genus 0; the projection $(t,H) \mapsto t$ is a double cover ramified at the zeros of Q_L. The degeneration loci in parameter space are controlled by the critical discriminant: Σ_L acquires a node precisely when $\Delta = 0$. The shadow growth rate ρ_L is the reciprocal of the distance from $t = 0$ to the nearest ramification point.

4. Main results

Theorem 4.1 (Main theorem: algebraicity, classification, and genus-2 formula)**.** *Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. Let $\kappa = \kappa(\mathcal{A})$ be the modular characteristic, and let α, S_4 be the cubic and quartic shadow invariants. Define the* shadow metric

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\,t + (9\alpha^2 + 16\kappa S_4)\,t^2$$

and the critical discriminant $\Delta = 8\kappa S_4$.

(i) ***Algebraicity.*** *The weighted shadow generating function* $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ *satisfies*

$$H(t)^2 = t^4\, Q(t).$$

Every shadow coefficient S_r *for* $r \geq 5$ *is determined by* (κ, α, S_4) *via* $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q(t)}$.

(ii) ***Shadow connection.*** *The shadow metric defines a logarithmic connection* $\nabla^{\mathrm{sh}} = d - Q'/(2Q)\, dt$ *with residue* $\frac{1}{2}$ *at each zero of* Q. *Monodromy* $= -1$ *(Koszul sign).*

(iii) ***Four-class partition.*** *Write* $Q = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$. *The shadow depth* $r_{\max}$ *lies in* $\{2, 3, 4, \infty\}$*:*
- *Class* **G***:* $\Delta = 0$, $\alpha = 0$, $r_{\max} = 2$.
- *Class* **L***:* $\Delta = 0$, $\alpha \neq 0$, $r_{\max} = 3$.
- *Class* **C***:* $\Delta = 0$ *on the primary line; quartic contact on a charged stratum,* $r_{\max} = 4$.
- *Class* **M***:* $\Delta \neq 0$, $r_{\max} = \infty$.

(iv) ***Universal factorization.*** *For* $r \geq 4$*:* $S_r = \Delta \cdot R_r$, *where* $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$ *is independent of the algebra.*

(v) ***Planted-forest formula.*** *The genus-2 planted-forest correction on a primary line is* $\delta^{(2,0)}_{\mathrm{pf}} = S_3(10\, S_3 - \kappa)/48$. *It vanishes for class* **G** *(where* $S_3 = 0$*).*

Proof. Parts (i)–(ii) are proved in Theorem 3.1 and Proposition 3.5, respectively. Part (iii) follows from Theorem 3.7. Part (iv) is the universal factorization in Theorem 3.7, equation (3.8). Part (v) is proved by explicit computation in Section 7. □

5. The Four Shadow Classes

The single-line dichotomy (Theorem 3.7) partitions the space of modular Koszul algebras into four shadow classes, determined by two invariants of the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$: the cubic coefficient $\alpha = S_3$ and the critical discriminant $\Delta = 8\kappa S_4$.

Definition 5.1 (Shadow classes)**.** Let $\mathcal{A}$ be a modular Koszul algebra with shadow metric Q_L on a one-dimensional primary line L. The *shadow class* of $\mathcal{A}$ along L is:

Class	α	Δ	$r_{\max}$	Q_L
G (Gaussian)	0	0	2	$(2\kappa)^2$
L (Lie/tree)	$\neq 0$	0	3	$(2\kappa + 3\alpha t)^2$
C (Contact)	0	0^*	4	$(2\kappa)^2$ (primary line)
M (Mixed)	$\neq 0$	$\neq 0$	∞	irreducible quadratic

Here $r_{\max}$ is the *shadow depth*: the largest arity r for which the shadow coefficient $S_r \neq 0$. *Class C has $\Delta = 0$ on the primary line (where the

single-line dichotomy applies), but the quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on a charged stratum; see Definition 5.4.

The classification is exhaustive: every entry in Table 1 falls into exactly one class, and every class is realized by at least one standard family. The mechanism is algebraic: the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ is algebraic of degree 2 (Theorem 3.1), and its Taylor coefficients S_r are computed from the convolution recursion $f^2 = Q_L$ with $f = \sum_{r\geq 2} S_r\, t^{r-2}$. When Q_L is a perfect square ($\Delta = 0$), the square root terminates; when Q_L is irreducible ($\Delta \neq 0$), the Taylor expansion is infinite.

5.1. **Class G: Gaussian ($r_{\max} = 2$).**

Definition 5.2. A modular Koszul algebra $\mathcal{A}$ is *Gaussian* if $S_r(\mathcal{A}) = 0$ for all $r \geq 3$. Equivalently, $\alpha = 0$ and $\Delta = 0$, so the shadow metric $Q_L(t) = 4\kappa^2$ is constant and the shadow tower terminates at the scalar curvature.

Examples. Heisenberg $\mathcal{H}_k$ with $\kappa = k$; lattice VOAs V_Λ with $\kappa = \mathrm{rank}(\Lambda)$; free fermion ψ with $\kappa = 1/4$; bc ghosts with $\kappa = -(6\lambda^2 - 6\lambda + 1)$.
Characterization. Class G algebras are freely generated as chiral algebras. Their bar complex is quadratic (the differential has no higher-order terms), and the OPE on each primary line is abelian. The full obstruction is captured by a single scalar:

$$\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda,$$

where η is the cyclic direction and $\Lambda \in H^*(\bar{\mathcal{M}}_g)$ is the tautological class. All genus-g data is controlled by $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$.

5.2. **Class L: Lie/tree ($r_{\max} = 3$).**

Definition 5.3. A modular Koszul algebra $\mathcal{A}$ is *Lie/tree* if $\alpha \neq 0$ and $\Delta = 0$, so that $S_3 \neq 0$ but $S_r = 0$ for all $r \geq 4$. The shadow metric is a perfect square: $Q_L(t) = (2\kappa + 3\alpha t)^2$.

Examples. Affine Kac–Moody $V_k(\mathfrak{g})$ for every simple Lie algebra $\mathfrak{g}$ and every non-critical level $k \neq -h^\vee$: type A ($\mathfrak{sl}_N$), types B, C, D (orthogonal and symplectic), exceptional types E_6, E_7, E_8, G_2, F_4. For all of these:

$$\kappa\big(V_k(\mathfrak{g})\big) = \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}\,.$$

Characterization. The nonzero cubic shadow $S_3 = \alpha$ arises from the Lie bracket (structure constants f_{abc}), while the Jacobi identity forces the quartic shadow $S_4 = 0$ on the primary line. The r-matrix $r(z) = \Omega_{\mathfrak{g}}/z$ (single simple pole; the bar construction absorbs one power of the OPE pole) encodes all nonlinear structure. The tree-level graph sum terminates at trivalent vertices.

5.3. **Class C: Contact ($r_{\max} = 4$).**

Definition 5.4. A modular Koszul algebra $\mathcal{A}$ is *contact* if $\Delta = 0$ on every primary line (so the single-line dichotomy gives $r_{\max}|_L \leq 3$ on each line), but there exists a charged stratum direction with $S_4 \neq 0$ (the quartic contact invariant), and the tower terminates at arity 4: $S_r = 0$ for $r \geq 5$.

Mechanism. Class C escapes the single-line dichotomy (which gives $r_{\max} \in \{2, 3, \infty\}$) via *stratum separation*: the quartic contact invariant Q^{contact} lives on a charged stratum whose self-bracket exits the cyclic deformation complex at arity 5, by rank-one rigidity. On the weight-changing primary line, the tower is abelian ($\alpha = 0$, $S_4 = 0$); the quartic contact is visible only on the full two-dimensional deformation space.
Examples. The $\beta\gamma$ system at conformal weight λ, with

$$\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1 = \tfrac{1}{2}\, c(\beta\gamma_\lambda),$$

and its Koszul dual, the bc system at the same weight. At the standard weight $\lambda = 0$ or $\lambda = 1$: $c = 2$ and $\kappa = 1$.

5.4. **Class M: Mixed ($r_{\max} = \infty$).**

Definition 5.5. A modular Koszul algebra $\mathcal{A}$ is *mixed* if $\alpha \neq 0$ and $\Delta \neq 0$, so the shadow metric Q_L is an irreducible quadratic in t and the shadow tower does not terminate.

Examples. Virasoro Vir_c with $\kappa = c/2$; $\mathcal{W}$-algebras $\mathcal{W}_N$ with $\kappa = (H_N - 1)\, c$ where $H_N = \sum_{j=1}^{N} 1/j$; Drinfeld–Sokolov reductions of affine Kac–Moody algebras.
Characterization. The shadow growth rate

$$\rho(\mathcal{A}) = \frac{\sqrt{9\alpha^2 + 16\kappa S_4}}{2|\kappa|}$$

controls the asymptotic behaviour of S_r: the coefficients satisfy $S_r \sim C(\mathcal{A})\, \rho(\mathcal{A})^r\, r^{-5/2} \cos(r\theta + \phi)$ for a branch-point argument θ determined by the shadow metric. The critical central charge c^* where $\rho(\mathrm{Vir}_c) = 1$ is the unique positive real root of the cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \qquad c^* \approx 6.125\,. \tag{5.1}$$

For $c > c^*$ the tower converges ($\rho < 1$), and for $c < c^*$ it diverges ($\rho > 1$). At the self-dual point $c = 13$: $\rho(\mathrm{Vir}_{13}) \approx 0.467$.

The Virasoro shadow tower has explicitly computed coefficients through arity 10:

$$S_2 = \tfrac{c}{2}\,, \qquad S_3 = 2\,, \qquad S_4 = \frac{10}{c(5c+22)}\,, \tag{5.2}$$

$$S_5 = \frac{-48}{c^2(5c+22)}\,, \quad S_6 = \frac{80(45c+193)}{3c^3(5c+22)^2}\,, \quad S_7 = \frac{-2880(15c+61)}{7c^4(5c+22)^2}\,. \tag{5.3}$$

Remark 5.6 (Depth and Koszulness). Shadow depth $r_{\max}$ does *not* characterize chiral Koszulness. All four classes—Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$)—consist of chirally Koszul algebras. Shadow depth classifies *complexity* within the Koszul world: the operadic complexity conjecture identifies $r_{\max}(\mathcal{A})$ with the A_∞-depth and L_∞-formality level.

Remark 5.7 (Total depth decomposition). The total shadow depth decomposes as $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ (Theorem 3.11), where the algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ and depths ≥ 5 are purely arithmetic (cusp forms contribute beyond the algebraic engine).

6. Shadow Tables

All entries in the tables below are verified against the compute layer. The modular characteristic $\kappa(\mathcal{A})$ is the arity-2 projection of the universal MC element $\Theta_{\mathcal{A}}$, and the genus-g free energy is $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ for single-generator (uniform-weight) algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ are the Faber–Pandharipande intersection numbers.

6.1. **Master shadow table.** Table 1 records, for each standard family, the modular characteristic κ, the shadow class, the shadow depth $r_{\max}$, the cubic coefficient $\alpha = S_3$, the quartic coefficient S_4, the critical discriminant $\Delta = 8\kappa S_4$.
†On the primary line, $S_4 = 0$ and $\Delta = 0$ for $\beta\gamma$; the quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on a charged stratum (stratum separation, Definition 5.4). In the table, $H_N = 1 + 1/2 + \cdots + 1/N$ denotes the N-th harmonic number. The Virasoro quartic contact $S_4 = Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c + 22)]$ is the unique quartic shadow on the T-line, proved by direct computation from the Virasoro OPE. The T-line shadow data ($S_3 = 2$, $S_4 = 10/[c(5c + 22)]$) is *universal*: it coincides for $\mathcal{W}_N$ at all $N \geq 2$, since every $\mathcal{W}_N$ contains the Virasoro subalgebra along the stress-tensor primary line. The additional primary lines (spin-3, spin-4, etc.) carry independent shadow data with separate quartic coefficients.

6.2. **Exceptional and non-simply-laced types.** Table 2 records the affine Kac–Moody data for exceptional and non-simply-laced types. All are class L. The kappa formula is universal: $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})\,(k + h^\vee)/(2h^\vee)$. The central charge sum $c + c' = 2\dim(\mathfrak{g})$ under the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$, and the complementarity $\kappa + \kappa' = 0$ holds for all affine Kac–Moody algebras. For non-simply-laced types, the dual Coxeter number $h^\vee$ differs from the Coxeter number h ($h^\vee = 3$ for B_2 and C_2, while $h = 4$). The shadow class is L regardless: the Jacobi identity kills the quartic on every primary line.

TABLE 1. Master shadow table: standard landscape

Algebra $\mathcal{A}$	$\boldsymbol{\kappa(\mathcal{A})}$	**Class**	$\boldsymbol{r_{\max}}$	$\boldsymbol{S_3}$	$\boldsymbol{S_4}$	$\boldsymbol{\Delta}$
Class G (Gaussian)						
Heisenberg $\mathcal{H}_k$	k	G	2	0	0	0
Lattice V_Λ	$\mathrm{rank}(\Lambda)$	G	2	0	0	0
Free fermion ψ	$1/4$	G	2	0	0	0
bc ghosts (weight λ)	$-(6\lambda^2-6\lambda+1)$	G	2	0	0	0
Class L (Lie/tree)						
$\widehat{\mathfrak{sl}}_2$ at level k	$3(k+2)/4$	L	3	$\neq 0$	0	0
$\widehat{\mathfrak{sl}}_N$ at level k	$(N^2-1)(k+N)/(2N)$	L	3	$\neq 0$	0	0
$\widehat{E}_6$ at level k	$13(k+12)/4$	L	3	$\neq 0$	0	0
$\widehat{E}_7$ at level k	$133(k+18)/36$	L	3	$\neq 0$	0	0
$\widehat{E}_8$ at level k	$62(k+30)/15$	L	3	$\neq 0$	0	0
Class C (Contact)						
$\beta\gamma$ (weight λ)	$6\lambda^2-6\lambda+1$	C	4	0	$0^\dagger$	$0^\dagger$
Class M (Mixed)						
Vir_c	$c/2$	M	∞	2	$\dfrac{10}{c(5c+22)}$	$\dfrac{40}{5c+22}$
$\mathcal{W}_3$ at c	$5c/6$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_4$ at c	$13c/12$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_5$ at c	$77c/60$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_N$ at c (general)	$(H_N-1)\,c$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$

Remark 6.1 (B_2/C_2 isomorphism). The Lie algebras $B_2 = \mathfrak{so}_5$ and $C_2 = \mathfrak{sp}_4$ are isomorphic, sharing dim $= 10$, $h^\vee = 3$, and $h = 4$. Their Cartan matrices are transposes of each other, but all shadow invariants coincide.

6.3. **Complementarity table.** ‡The Koszul dual $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ is not a Heisenberg algebra (same κ, different OPE structure); the central charge sum is not defined within a single-parameter family.
Here $K_N = 2(N-1) + 4(N^2-1)N$ is the Koszul conductor for $\mathcal{W}_N = \mathcal{W}(\mathfrak{sl}_N)$ (the sum $c + c'$ under the Feigin–Frenkel involution $k \mapsto -k - 2N$ on the underlying $\mathfrak{sl}_N$). The key structural distinction: for all affine Kac–Moody algebras, $\kappa + \kappa' = 0$ (anti-symmetry under the Feigin–Frenkel involution); for $\mathcal{W}$-algebras, $\kappa + \kappa' = (H_N - 1) \cdot K_N \neq 0$. The Virasoro ($N = 2$) gives $\kappa + \kappa' = \frac{1}{2} \cdot 26 = 13$, a nonzero constant independent of c. The Koszul

TABLE 2. Shadow data for exceptional and non-simply-laced types

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\boldsymbol{\kappa(V_k(\mathfrak{g}))}$	$c(V_k)+c(V_{k'})$	**Class**	$r_{\max}$
$B_2 \cong \mathfrak{so}_5$	10	3	$5(k{+}3)/3$	20	L	3
$C_2 \cong \mathfrak{sp}_4$	10	3	$5(k{+}3)/3$	20	L	3
G_2	14	4	$7(k{+}4)/4$	28	L	3
F_4	52	9	$26(k{+}9)/9$	104	L	3
E_6	78	12	$13(k{+}12)/4$	156	L	3
E_7	133	18	$133(k{+}18)/36$	266	L	3
E_8	248	30	$62(k{+}30)/15$	496	L	3

TABLE 3. Complementarity data: $\kappa(\mathcal{A})+\kappa(\mathcal{A}^!)$ and $c(\mathcal{A})+c(\mathcal{A}^!)$ under Koszul duality

Family $\mathcal{A}$	**Koszul dual $\mathcal{A}^!$**	$\boldsymbol{c+c'}$	$\boldsymbol{\kappa+\kappa'}$	**Self-dual point**
$\mathcal{H}_k$	$\mathrm{Sym}^{\mathrm{ch}}(V^*)$	‡	0	none
V_Λ (lattice)	$V_{\Lambda^!}$	‡	0	Λ self-dual
Free fermion	$\mathrm{Sym}^{\mathrm{ch}}(\gamma)$	0	0	—
$\beta\gamma_\lambda$	bc_λ	0	0	—
$V_k(\mathfrak{sl}_2)$	$V_{-k-4}(\mathfrak{sl}_2)$	6	0	$k=-2$ (critical)
$V_k(\mathfrak{sl}_N)$	$V_{-k-2N}(\mathfrak{sl}_N)$	$2(N^2{-}1)$	0	$k=-N$ (critical)
$V_k(\mathfrak{g})$ (general)	$V_{-k-2h^\vee}(\mathfrak{g})$	$2\dim(\mathfrak{g})$	0	$k=-h^\vee$ (critical)
Vir_c	Vir_{26-c}	26	13	$c=13$
$\mathcal{W}_3$ at c	$\mathcal{W}_3$ at $100{-}c$	100	250/3	$c=50$
$\mathcal{W}_4$ at c	$\mathcal{W}_4$ at $246{-}c$	246	533/2	$c=123$
$\mathcal{W}_5$ at c	$\mathcal{W}_5$ at $488{-}c$	488	9394/15	$c=244$
$\mathcal{W}_N$ at c (general)	$\mathcal{W}_N$ at $K_N{-}c$	K_N	$(H_N{-}1)\,K_N$	$c=K_N/2$

involution $c\mapsto 26-c$ has a fixed point at $c=13$, where $\kappa(\mathrm{Vir}_{13})=13/2$ and Vir_{13} is self-dual under uncurved quadratic duality.

6.4. **Genus tables.** The Faber–Pandharipande numbers and the genus expansion $F_g(\mathcal{A})=\kappa(\mathcal{A})\cdot\lambda_g^{\mathrm{FP}}$.
The formula is $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}}\cdot\frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$-th Bernoulli number. The generating function is $\sum_{g\geq 1}\lambda_g^{\mathrm{FP}}\,\hbar^{2g} = \hat{A}(i\hbar)-1$, where

TABLE 4. Faber–Pandharipande intersection numbers λ_g^{FP}

g	λ_g^{FP}	Decimal
1	1/24	$0.041\overline{6}$
2	7/5760	1.215×10^{-3}
3	31/967680	3.204×10^{-5}
4	127/154828800	8.201×10^{-7}
5	73/3503554560	2.084×10^{-8}

$\hat{A}(x) = (x/2)/\sinh(x/2)$ is the $\hat{A}$-genus (so $\hat{A}(ix) = (x/2)/\sin(x/2)$). The numbers λ_g^{FP} are strictly positive, strictly decreasing, and decay like $(2\pi)^{-2g}$.

TABLE 5. Genus expansion $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\mathrm{FP}}$ for selected families

Algebra $\mathcal{A}$	$\boldsymbol{\kappa}$	$\boldsymbol{F_1}$	$\boldsymbol{F_2}$	$\boldsymbol{F_3}$
$\mathcal{H}_1$ (Heisenberg, k=1)	1	1/24	7/5760	31/967680
$V_1(\mathfrak{sl}_2)$	9/4	3/32	7/2560	31/430080
$V_1(\mathfrak{sl}_3)$	16/3	2/9	7/1080	31/181440
V_Λ (Leech, rank =24)	24	1	7/240	31/40320
$\beta\gamma$ (standard, λ=1)	1	1/24	7/5760	31/967680
Vir_1	1/2	1/48	7/11520	31/1935360
Vir_{13}	13/2	13/48	91/11520	403/1935360
Vir_{26}	13	13/24	91/5760	403/967680
$\mathcal{W}_3$ at c=6	5	5/24	7/1152	31/193536

The entries for $V_1(\mathfrak{sl}_3)$ use $\kappa = (8 \cdot 4)/(2 \cdot 3) = 16/3$. The Leech lattice (rank = 24) gives $F_1 = 1$, reflecting the Jacobi Δ-function origin of the Leech theta series.

6.5. **Shadow growth rate table.** The shadow growth rate $\rho = \sqrt{(9\alpha^2 + 16\kappa S_4)/(4\kappa^2)}$ determines the convergence of the shadow tower. For Virasoro, $\rho^2 = (180c + 872)/[c^2(5c + 22)]$. The critical central charge $c^* \approx 6.125$ (the positive real root of (5.1)) separates the convergent regime ($c > c^*$, tower summable) from the divergent regime ($c < c^*$, tower grows exponentially). Class G and class L algebras have $\rho = 0$ (the tower terminates, so there is no exponential growth).

6.6. **Virasoro shadow tower table.** Each S_r is a rational function of c with poles at $c = 0$ and $c = -22/5$; these are the two values where the quadratic Casimir $c(5c+22)$ vanishes. The sign pattern $S_r > 0$ for even r and

TABLE 6. Shadow growth rate $\rho(\mathcal{A})$ for class M algebras (Virasoro at selected central charges)

c	κ	Q^{contact}	ρ	**Regime**
1/2 (Ising)	1/4	40/49	≈ 12.53	divergent
1	1/2	10/27	≈ 6.24	divergent
4	2	5/84	≈ 1.54	divergent
$c^* \approx 6.125$	≈ 3.06	≈ 0.031	1	critical
13 (self-dual)	13/2	10/1131	≈ 0.467	convergent
25	25/2	2/735	≈ 0.242	convergent
26 (bosonic string)	13	5/1976	≈ 0.232	convergent

TABLE 7. Virasoro shadow coefficients S_r for $r = 2, \ldots, 7$

r	$S_r(\mathrm{Vir}_c)$
2	$c/2$
3	2
4	$10/[c(5c+22)]$
5	$-48/[c^2(5c+22)]$
6	$80(45c+193)/[3c^3(5c+22)^2]$
7	$-2880(15c+61)/[7c^4(5c+22)^2]$
8	$80(2025c^2+16470c+33314)/[c^5(5c+22)^3]$
9	$-1280(2025c^2+15570c+29554)/[3c^6(5c+22)^3]$
10	$256(91125c^3+1050975c^2+3989790c+4969967)/[c^7(5c+22)^4]$

$S_r < 0$ for odd $r \geq 5$ (at $c > 0$) reflects the oscillatory decay $\cos(r\theta + \phi)$ from the complex branch points of the shadow generating function. All entries have been verified by two independent methods: closed-form derivation and the convolution recursion from $f^2 = Q_L$.

7. The genus-2 free energy of W_3

The preceding sections developed the theory in the single-channel regime: one strong generator, one primary line, one modular characteristic κ. We now compute the first explicit multi-channel invariant. The W_3 algebra has two strong generators—the stress tensor T of conformal weight 2 and the spin-3 current W of conformal weight 3—and the genus-2 free energy $F_2(W_3)$ receives contributions from *both* channels and their interactions.

7.1. **The Frobenius algebra of the bar-complex CohFT.** The bar-complex CohFT of W_3 has a two-dimensional underlying Frobenius algebra with basis $\{e_T, e_W\}$. The metric is diagonal, determined by the per-channel modular characteristics:

$$\eta_{TT} = \kappa_T = \frac{c}{2}, \qquad \eta_{WW} = \kappa_W = \frac{c}{3}, \qquad \eta_{TW} = 0. \tag{7.1}$$

The total modular characteristic is $\kappa(W_3) = \kappa_T + \kappa_W = 5c/6 = c \cdot (H_3 - 1)$, where $H_3 = 1 + 1/2 + 1/3 = 11/6$ is the third harmonic number. The inverse metric gives the propagators:

$$\eta^{TT} = \frac{2}{c}, \qquad \eta^{WW} = \frac{3}{c}. \tag{7.2}$$

Both propagators have weight 1: the bar-complex propagator is $d\log E(z,w)$, which has weight 1 regardless of the conformal weight of the field being sewed.

The three-point structure constants are determined by the OPE. The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ forces all three-point functions with an odd number of W-insertions to vanish:

$$C_{TTT} = C_{TWW} = C_{WWT} = c, \qquad C_{TTW} = C_{TWT} = C_{WTT} = C_{WWW} = 0. \tag{7.3}$$

The derivations are: $T_{(1)}T = 2T$ gives $C^T_{TT} = 2$, whence $C_{TTT} = \eta_{TT} \cdot 2 = c$; $T_{(1)}W = 3W$ gives $C^W_{TW} = 3$, whence $C_{TWW} = \eta_{WW} \cdot 3 = c$; $W_{(3)}W = 2T$ gives $C^T_{WW} = 2$, whence $C_{WWT} = \eta_{TT} \cdot 2 = c$.

The genus-0 four-point vertex is universal. For any channel pair $(i,j) \in \{T,W\}^2$:

$$V_{0,4}(i,i \mid j,j) \;=\; \sum_m \eta^{mm}\, C_{iim}\, C_{jjm} \;=\; \eta^{TT} \cdot c \cdot c \;=\; 2c. \tag{7.4}$$

Only the T-channel contributes to the sum: $C_{iiT} = c$ for $i \in \{T,W\}$, while $C_{iiW} = 0$ for both $i = T$ (since $C_{TTW} = 0$) and $i = W$ (since $C_{WWW} = 0$). The W-channel is invisible at genus-0 valence 4.

7.2. **The six stable graphs at genus 2.** The moduli space $\overline{\mathcal{M}}_{2,0}$ has a stratification by six stable graphs $\Gamma_0, \ldots, \Gamma_5$:

	Name	**Vertices**	**Edges**	$\lvert\mathrm{Aut}\rvert$	h^1	**Type**
Γ_0	smooth	(g=2, val=0)	0	1	0	compact stratum
Γ_1	figure-eight	(g=1, val=2)	1 self-loop	2	1	one-loop
Γ_2	banana	(g=0, val=4)	2 self-loops	8	2	two-loop sunset
Γ_3	dumbbell	(g=1, val=1) + (g=1, val=1)	1 bridge	2	0	separating
Γ_4	theta	(g=0, val=3) + (g=0, val=3)	3 bridges	12	2	non-separating
Γ_5	lollipop	(g=0, val=3) + (g=1, val=1)	1 self-loop + 1 bridge	2	1	mixed

Every vertex satisfies the stability condition $2g_v + \mathrm{val}_v \geq 3$, and the genus formula $h^1(\Gamma) + \sum_v g_v = 2$ holds in each case. Note that the smooth graph Γ_0 has no edges and contributes the integrated class on $\mathcal{M}_{2,0}$ itself; the remaining five graphs are the boundary strata.

7.3. **Multi-channel Feynman rules.** For each graph Γ with $|E(\Gamma)|$ edges, a *channel assignment* is a function $\sigma\colon E(\Gamma) \to \{T, W\}$. The Feynman rules of the bar-complex CohFT assign to each pair (Γ, σ) an amplitude

$$\mathcal{A}(\Gamma, \sigma) \;=\; \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e),\,\sigma(e)} \prod_{v \in V(\Gamma)} V_v\big(g_v;\, \sigma|_{\text{half-edges}(v)}\big), \tag{7.5}$$

where the vertex factors are:

- *Genus-0, trivalent*: $V_v = C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.
- *Genus-0, four-valent* (the banana vertex): $V_v = \sum_m \eta^{mm} C_{\sigma(e_1)\sigma(e_2)m}\, C_{m\,\sigma(e_3)\sigma(e_4)}$.
- *Genus-1, valence n*: $V_v = \kappa_{\sigma(e)}/24$ (per-channel genus-1 universality, proved unconditionally).

Each half-edge at a self-loop contributes a pair of half-edges to the same vertex, both carrying channel $\sigma(e)$.

7.4. **Per-graph amplitudes.** We now compute the amplitude of each boundary graph, decomposed into *diagonal* (all edges carry the same channel) and *mixed* (at least two edges carry different channels) contributions.

Γ_1*: figure-eight.* One edge, two channel assignments. The vertex is genus-1 with valence 2, so the vertex factor for channel i is $\kappa_i/24$:

$$\mathcal{A}(\Gamma_1, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \tfrac{\kappa_T}{24} = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c}{48} = \tfrac{1}{48}, \tag{7.6}$$

$$\mathcal{A}(\Gamma_1, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \tfrac{\kappa_W}{24} = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c}{72} = \tfrac{1}{48}. \tag{7.7}$$

Both channels contribute equally: the propagator $1/\kappa_i$ cancels the genus-1 vertex factor $\kappa_i/24$, leaving the universal value $1/48$ independent of c. The total is $1/24 = \lambda_1^{\mathrm{FP}}$. No mixed channel assignments exist (single edge).

Γ_2*: banana.* Two edges, four channel assignments, $|\mathrm{Aut}| = 8$. The vertex is genus-0 with valence 4, and the four-point vertex factor is $V_{0,4}(i, i \mid j, j) = 2c$ universally (7.4).

$$\mathcal{A}(\Gamma_2, TT) = \tfrac{1}{8}\,(\eta^{TT})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{4}{c^2} \cdot 2c = \frac{1}{c}, \tag{7.8}$$

$$\mathcal{A}(\Gamma_2, WW) = \tfrac{1}{8}\,(\eta^{WW})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{9}{c^2} \cdot 2c = \frac{9}{4c}, \tag{7.9}$$

$$\mathcal{A}(\Gamma_2, TW{+}WT) = \tfrac{1}{8} \cdot 2 \cdot \eta^{TT}\,\eta^{WW} \cdot 2c = \tfrac{1}{8} \cdot 2 \cdot \tfrac{6}{c^2} \cdot 2c = \frac{3}{c}. \tag{7.10}$$

The cross-channel contribution is $\delta_{\text{banana}} = 3/c$.

Γ_3*: dumbbell.* One bridge edge, two channel assignments, $|\mathrm{Aut}| = 2$. Each vertex has genus 1 and valence 1, with vertex factor $\kappa_i/24$:

$$\mathcal{A}(\Gamma_3, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \left(\tfrac{\kappa_T}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c^2}{2304} = \frac{c}{2304}, \tag{7.11}$$

$$\mathcal{A}(\Gamma_3, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \left(\tfrac{\kappa_W}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c^2}{5184} = \frac{c}{3456}. \tag{7.12}$$

The total is $c/2304 + c/3456 = (3c + 2c)/6912 = 5c/6912 = \kappa(W_3)/1152$. No mixed channel assignments exist (single edge).

Γ_4*: theta.* Three bridge edges, eight channel assignments, $|\mathrm{Aut}| = 12$. Each vertex has genus 0 and valence 3, with vertex factor $C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.

- *All-T*: both vertices get $C_{TTT} = c$. Amplitude $= \frac{1}{12}\,(\eta^{TT})^3 \cdot c^2 = \frac{1}{12} \cdot \frac{8}{c^3} \cdot c^2 = \frac{2}{3c}$.
- *All-W*: both vertices get $C_{WWW} = 0$. Amplitude $= 0$.
- *Type* (T, T, W) (three such assignments): each vertex receives one W half-edge, giving $C_{TTW} = 0$. All three vanish.
- *Type* (T, W, W) (three such assignments): each vertex receives half-edges (T, W, W), giving $C_{TWW} = c$. The propagator product is $\eta^{TT}(\eta^{WW})^2 = (2/c)(9/c^2) = 18/c^3$. Each assignment contributes $18c^2/c^3 = 18/c$; with $|\mathrm{Aut}| = 12$, the total mixed amplitude is

$$\delta_{\text{theta}} = \frac{3 \cdot 18}{12c} = \frac{9}{2c}. \tag{7.13}$$

Γ_5*: lollipop.* One self-loop at vertex v_0 (genus 0, valence 3) and one bridge to vertex v_1 (genus 1, valence 1). Two edges, four channel assignments, $|\mathrm{Aut}| = 2$.

- (T, T): self-loop T, bridge T. Vertex v_0 gets half-edges $(T, T, T) \to C_{TTT} = c$. Vertex v_1 gets half-edge $T \to \kappa_T/24 = c/48$. Propagators: $(\eta^{TT})^2 = 4/c^2$. With $|\mathrm{Aut}|$: $\frac{1}{2} \cdot \frac{4}{c^2} \cdot c \cdot \frac{c}{48} = \frac{1}{24}$.
- (W, W): self-loop W, bridge W. Vertex v_0 gets $(W, W, W) \to C_{WWW} = 0$. Vanishes.
- (T, W): self-loop T, bridge W. Vertex v_0 gets $(T, T, W) \to C_{TTW} = 0$. Vanishes.
- (W, T): self-loop W, bridge T. Vertex v_0 gets $(W, W, T) \to C_{WWT} = c$. Vertex v_1 gets $T \to \kappa_T/24 = c/48$. Propagators: $\eta^{WW} \cdot \eta^{TT} = (3/c)(2/c) = 6/c^2$. With $|\mathrm{Aut}|$:

$$\delta_{\text{lollipop}} = \tfrac{1}{2} \cdot \tfrac{6}{c^2} \cdot c \cdot \tfrac{c}{48} = \tfrac{6c^2}{96c^2} = \frac{1}{16}. \tag{7.14}$$

The lollipop mixed amplitude $1/16$ is *independent of* c: the c^2 in the numerator (from $C_{WWT} \cdot \kappa_T/24$) cancels the c^2 in the denominator (from $\eta^{WW} \cdot \eta^{TT}$).

7.5. **The cross-channel correction.** Collecting the mixed-channel amplitudes from all boundary graphs:

$$\delta F_2 \;=\; \underbrace{\frac{3}{c}}_{\text{banana}} + \underbrace{\frac{9}{2c}}_{\text{theta}} + \underbrace{\frac{1}{16}}_{\text{lollipop}} \;=\; \frac{15}{2c} + \frac{1}{16} \;=\; \frac{c+120}{16c}. \tag{7.15}$$

The figure-eight and dumbbell graphs contribute zero mixed amplitude (single edges admit only diagonal assignments). The correction decomposes into two structurally distinct pieces:

- A *c-dependent* piece $15/(2c)$, arising from genus-0 vertex factors (the banana and theta graphs). This is R-matrix independent: genus-0 vertices receive no corrections from the CohFT R-matrix.
- A *c-independent* piece $1/16$, arising from the lollipop graph, which mixes a genus-0 vertex factor with a genus-1 vertex factor $\kappa_i/24$.

Remark 7.1 (Comparison with the scalar approximation). The per-channel (diagonal) graph sum, computed by applying Theorem D to each channel independently, gives

$$F_2^{\mathrm{scal}}(W_3) = \kappa \cdot \lambda_2^{\mathrm{FP}} = \frac{5c}{6} \cdot \frac{7}{5760} = \frac{7c}{6912}. \tag{7.16}$$

The ratio of the naive cross-channel correction to the scalar approximation is

$$\frac{\delta F_2}{F_2^{\mathrm{scal}}} = \frac{432(c+120)}{7c^2}, \tag{7.17}$$

which diverges as $c \to 0$ and decays as c^{-1} at large c. At $c = 4$: the ratio is approximately 478, and the cross-channel correction *dominates* the scalar term. At $c = 50$: the ratio is approximately 4.2. At large c, the lollipop's c-independent contribution $1/16$ dominates, and the correction becomes a subleading $O(1/c)$ effect relative to the $O(c)$ scalar term.

Remark 7.2 (Status of the cross-channel correction). The computation above uses the naive genus-1 vertex factors $\kappa_i/24$ without R-matrix corrections. Whether the CohFT R-matrix produces compensating terms that cancel δF_2 is the content of the multi-generator universality problem (**op:multi-generator-universality**). If universality holds, the R-matrix corrections at genus 1 must produce exactly $-(c+120)/(16c)$, a nontrivial identity on the CohFT structure. The computation stands as an explicit prediction that any proof of multi-generator universality must reproduce.

7.6. **The complete amplitude table.** We record the full per-graph data:

Graph	**Diagonal (T)**	**Diagonal (W)**	**Mixed**	**Total**
Γ_1 figure-eight	$1/48$	$1/48$	0	$1/24$
Γ_2 banana	$1/c$	$9/(4c)$	$3/c$	$25/(4c)$
Γ_3 dumbbell	$c/2304$	$c/3456$	0	$5c/6912$
Γ_4 theta	$2/(3c)$	0	$9/(2c)$	$31/(6c)$
Γ_5 lollipop	$1/24$	0	$1/16$	$5/48$

The table records the boundary-graph amplitudes only; the smooth graph Γ_0 contributes an additional integrated class on $\mathcal{M}_{2,0}$ whose per-channel decomposition is determined by the constraint $\sum_\Gamma \mathcal{A}(\Gamma) = \kappa \cdot \lambda_2^{\mathrm{FP}}$ when the full universality identity holds. The mixed column sums to $\delta F_2 = (c+120)/(16c)$.

Every entry in this table has been verified against the compute module `w3_genus2.py`, using exact rational arithmetic (`fractions.Fraction`), at the seven test central charges $c \in \{1, 2, 4, 13, 26, 50, 100\}$.

7.7. **Planted-forest corrections.** The planted-forest correction on each primary line is $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$.

T-line: $S_3^T = 2$ (the Virasoro cubic shadow), $\kappa_T = c/2$.

$$\delta_{\mathrm{pf}}^T = \frac{2(20 - c/2)}{48} = \frac{40 - c}{48}. \tag{7.18}$$

This vanishes at $c = 40$, is positive for $c < 40$, and negative for $c > 40$.

W-line: $S_3^W = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$, which forces all odd-arity shadows on the W-line to vanish. Hence $\delta_{\mathrm{pf}}^W = 0$.

The total planted-forest correction is $\delta_{\mathrm{pf}} = (40 - c)/48$, living entirely on the T-line.

7.8. **Koszul duality constraints.** Under the Koszul involution, W_3 at central charge c is dual to W_3 at central charge $c' = 100 - c$ (the Feigin–Frenkel involution for $\mathfrak{sl}_3$). The modular characteristics satisfy

$$\kappa(c) + \kappa(100 - c) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{250}{3}, \tag{7.19}$$

and the scalar free energies satisfy $F_2^{\mathrm{scal}}(c) + F_2^{\mathrm{scal}}(100 - c) = (250/3) \cdot \lambda_2^{\mathrm{FP}}$. The cross-channel correction at the dual pair is

$$\delta F_2(c) + \delta F_2(100 - c) = \frac{c + 120}{16c} + \frac{220 - c}{16(100 - c)} = -\frac{c^2 - 100c - 6000}{8c(100 - c)}.$$

This sum is not constant in c: the cross-channel correction does not respect Koszul additivity. If universality holds (δF_2 is cancelled by R-matrix corrections), this is automatic. If the naive correction persists, it produces a Koszul-asymmetric contribution to F_2—an observable departure from single-channel behaviour.

8. The gravitational coproduct is primitive

In gauge theory, the gluon is a fundamental field: it can split. A single gluon entering a scattering region can exit as two gluons, and the coproduct of the dg-shifted Yangian encodes this splitting at the algebraic level. In gravity, the graviton is not fundamental but composite. The stress tensor is the Sugawara quadratic

$$T = \frac{1}{k+2}\Big(\tfrac{1}{2}{:}J^0J^0{:} + {:}J^+J^-{:}\Big), \tag{8.1}$$

built from two copies of the affine connection by normal ordering. The compositeness of the graviton has a precise algebraic consequence: the transferred coproduct is trivial.

8.1. **The DS-HPL transfer theorem.** The Drinfeld–Sokolov reduction $\hat{\mathfrak{sl}}_2 \to \mathrm{Vir}$ is a BRST reduction with a strong deformation retract (i, p, h) from the BRST complex to the Virasoro cohomology. The homotopy h has ghost-number degree -1: it maps $V \mapsto b$ (ghost number $0 \to -1$) and $c \mapsto U/2$ (ghost number $1 \to 0$).

Theorem 8.1 (DS-HPL transfer)**.** *Let $Y^{\mathrm{dg}}(\widehat{\mathfrak{sl}}_2)$ denote the dg-shifted Yangian at level k, with A_∞ products $\{m_n^{\mathrm{aff}}\}$, coproduct Δ_z^{aff}, and r-matrix $r^{\mathrm{aff}}(z) = \Omega/z$. The HPL transfer through the BRST SDR produces:*

(i) *Transferred A_∞ products $\{m_n^{\mathrm{Vir}}\}_{n\geq 2}$ on the Virasoro sector, with $m_n^{\mathrm{Vir}} \neq 0$ for all $n \geq 2$ (the entire infinite A_∞ tower of class $\mathbf{M}$).*
(ii) *A transferred coproduct $\Delta_z^{\mathrm{Vir}}\colon H_{\mathrm{Vir}} \to H_{\mathrm{Vir}} \otimes H_{\mathrm{Vir}}$.*
(iii) *A transferred r-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, with cubic leading pole reflecting the Sugawara nonlinearity.*

The r-matrix deserves comment. The affine r-matrix $r^{\mathrm{aff}}(z) = \Omega/z$ is a simple pole—the collision residue of the affine bar complex, whose OPE has at most a double pole. The Virasoro OPE has a quartic pole (z^{-4} in $T(z)T(w)$); the bar kernel $d\log(z-w)$ absorbs one power, producing the cubic pole z^{-3} of $r^{\mathrm{Vir}}(z)$. This is the first instance of the general principle: the r-matrix pole order equals the OPE pole order minus one.

Theorem 8.2 (Gravitational coproduct primitivity)**.** *The HPL-transferred coproduct is strictly primitive at all arities:*

$$\Delta_{z,k}^{\mathrm{Vir}} = 0 \qquad \textit{for all } k \geq 2. \tag{8.2}$$

That is, $\Delta_z^{\mathrm{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. All nontrivial structure of the Virasoro dg-shifted Yangian resides in the $r(z)$-twisted target product, never in the coproduct.

Proof. The proof is a ghost-number obstruction, proceeding in three steps.

Step 1. The linear transferred coproduct $(p\otimes p)(\Delta_z(T))$ is already primitive. The affine coproduct $\Delta_z(T)$ expands as $\tau_z(T) \otimes 1 + 1 \otimes T + \frac{1}{k+2}\,\Omega(w{+}z,\, w)$, where $\Omega(w{+}z, w) = J^0(w{+}z) \otimes J^0(w) + J^+(w{+}z) \otimes J^-(w) + J^-(w{+}z) \otimes$

$J^+(w)$. The projection p annihilates J^0 (not in the Virasoro sector) and J^+ decomposes as $V+1$ with $p(V)=0$. Every summand of Ω contains at least one factor from $\{J^0, J^-, J^+\}$ at conformal weight 1, where the Virasoro sector has no states ($L_{-1}|0\rangle = 0$). Each tensor factor of $(p\otimes p)(\Omega)$ has p killing at least one component, so $(p\otimes p)(\Omega)=0$.

Step 2. At arity $k\geq 2$, every HPL tree has at least one internal edge labelled by the homotopy h. The ghost-number accounting is:

Operation	Ghost degree
h (homotopy)	-1
δ (BRST perturbation)	$+1$
μ (vertex algebra product)	0
Δ_z (coproduct)	0

The product $h\delta$ has ghost degree $-1+1=0$, but the tree topology at arity $k\geq 2$ always contains one *uncompensated* h: the outermost homotopy that is not preceded by a δ insertion. The overall output before projection has ghost number ≤ -1.

Step 3. The W-algebra cohomology sits in ghost number 0. The projection p annihilates all ghost-number-($\neq 0$) elements. Since the output of each arity-k tree has ghost number ≤ -1, the projection $(p\otimes p)$ kills every summand: in each tensor factor, at least one component has negative ghost number.

The perturbation does not produce corrections. The BRST perturbation δ has ghost degree $+1$, so the product $h\delta$ preserves ghost number. But $\delta(T)=0$ (the stress tensor is BRST-closed), so $(h\delta)(i(T))=0$, and by induction $(h\delta)^n(i(T))=0$ for all $n\geq 1$. The corrected inclusion $i_\infty(T)=i(T)$ is unperturbed, and the same primitive coproduct results at every perturbative order. □

Conjecture 8.3 (General DS primitivity)**.** *For any simple Lie algebra $\mathfrak{g}$, the HPL-transferred coproduct of the principal $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g})$ is strictly primitive at all arities. The ghost-number obstruction of the BRST complex for the principal Drinfeld–Sokolov reduction $V_k(\mathfrak{g})\to\mathcal{W}_k(\mathfrak{g})$ forces $\Delta^{\mathcal{W}}_{z,n}=0$ for $n\geq 2$.*

8.2. **Physical meaning.** The coproduct Δ_z of a dg-shifted Yangian encodes the *splitting* of excitations. Its strictness or nontriviality is the algebraic fingerprint of the distinction between fundamental and composite fields.

Theory	**Boundary algebra**	$\boldsymbol{\Delta_z}$	**Shadow depth**	**Reason**
Chern–Simons	$\hat{\mathfrak{g}}_k$ (affine KM)	$\neq\varepsilon$	3 (class **L**)	fundamental gluons split
3d gravity	$\mathrm{Vir}_c = W_k(\mathfrak{sl}_2)$	$=\varepsilon$	∞ (class **M**)	ghost obstruction (DS)
Twisted holography	H_k (Heisenberg)	$=\varepsilon$	2 (class **G**)	free field, $r=k/z$
M2 brane	DDCA	$\neq\varepsilon$	—	no DS, fundamental modes
M5 brane	$W_{1+\infty}(K)$	$=\varepsilon$	∞ (class **M**)	ghost obstruction (DS)

The pattern is clean: *theories arising from principal Drinfeld–Sokolov reduction have primitive coproducts.* The ghost sector of the BRST complex provides the grading obstruction that kills all higher-arity coproduct components. Theorem 8.2 proves this for $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$; the same ghost-number argument applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$ (the ghost-number grading of the BRST complex is universal), yielding Conjecture 8.3 in the main text. Theories with fundamental excitations (Chern–Simons, deformed double current algebras) have no ghost sector available, and their coproducts are nontrivial.

The two exceptions to the DS pattern are illuminating. The Heisenberg algebra has a primitive coproduct for a different reason: it is a free field, its r-matrix $r^H(z) = k/z$ is scalar, and the bar complex is formal at arity 2 (shadow depth $r_{\max} = 2$, class $\mathbf{G}$). No DS reduction and no ghost obstruction; primitivity follows from the absence of nonlinear OPE structure. The M2 brane boundary algebra (a deformed double current algebra) has a nontrivial coproduct because it is not obtained by DS reduction from any affine algebra—its excitations are fundamental modes of the M2 worldvolume theory.

8.3. **The r-matrix carries all gravitational structure.** With the coproduct primitive, the full gravitational content of the Virasoro dg-shifted Yangian is encoded in the r-matrix

$$r^{\mathrm{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{8.3}$$

The two terms encode distinct physical data:

- The *leading coefficient* $c/2 = \kappa(\mathrm{Vir}_c)$ is the modular characteristic. It determines the genus tower $F_g = (c/2)\int_{\overline{\mathcal{M}}_g} \lambda_g$ at all genera, the BTZ black hole entropy $S_{\mathrm{BTZ}} = 2\pi\sqrt{ch/6}$, and the Hawking–Page transition at $h = c/24$.
- The *subleading term* $2T/z$ encodes the state dependence of gravitational scattering. The braiding of Wilson lines (gravitational line operators) in the holomorphic-topological bulk is controlled by this term.
- The *pole structure*—cubic leading pole, simple subleading pole, no even poles—is the signature of a bosonic algebra extracted through the $d\log$ kernel: the OPE poles z^{-4}, z^{-2}, z^{-1} become r-matrix poles z^{-3}, z^{-1} (each shifted by -1), with the z^{-2} OPE pole producing z^{-1} (which merges with the subleading term from z^{-1}).

The contrast with Chern–Simons is maximal. In Chern–Simons gauge theory, the affine r-matrix $r^{\mathrm{aff}}(z) = \Omega/z$ is a simple pole and the coproduct is nontrivial: gluons split and the splitting is nontrivial. In gravity, the r-matrix has a cubic pole and the coproduct is trivial: gravitons braid but do not split. The Drinfeld–Sokolov reduction bridges the two: it takes the affine Casimir Ω/z to the gravitational $(c/2)/z^3 + 2T/z$ by homological

perturbation through the Sugawara nonlinearity, and in the same stroke, the ghost-number obstruction kills the nontrivial coproduct.

Datum	**Chern–Simons**	**Gravity**
boundary algebra	$V_k(\mathfrak{g})$ (affine KM)	Vir_c (Virasoro)
highest OPE pole	z^{-2} (quadratic)	z^{-4} (quartic)
A_∞ tower	strict ($m_{k\geq 3} = 0$)	infinite ($m_k \neq 0$ all k)
shadow depth	3 (class **L**)	∞ (class **M**)
$r^{\mathrm{coll}}(z)$	Ω/z (simple pole)	$(c/2)/z^3 + 2T/z$ (cubic pole)
Δ_z	nontrivial (gluons split)	primitive (gravitons braid)
κ	$\dim \mathfrak{g}\,(k+h^\vee)/(2h^\vee)$	$c/2$
$\kappa + \kappa^!$	0	13

The entire table descends from a single datum: the number of λ-powers in the λ-bracket. Affine Kac–Moody has λ^1 at most (quadratic pole), yielding strict associativity, finite shadow depth, a simple-pole r-matrix, and a nontrivial coproduct. The Virasoro algebra has λ^3 (quartic pole), yielding the full infinite A_∞ tower, infinite shadow depth, a cubic-pole r-matrix, and a primitive coproduct. The Drinfeld–Sokolov reduction is the functor that connects the two columns: it increases the λ-bracket order by two, trades the nontrivial coproduct for a ghost obstruction, and replaces the complementarity $\kappa + \kappa^! = 0$ with $\kappa + \kappa^! = 13$.

In one sentence: *gravity does not produce particles, only braids; the graviton is composite, and the ghost sector of its composite origin is the algebraic obstruction that kills the splitting.*

9. Open problems and outlook

We collect several directions suggested by the results of this paper.

9.1. **Multi-generator universality at genus ≥ 2.** The genus-2 computation for $\mathcal{W}_3$ (Section 7) produces a cross-channel correction $\delta F_2 = (c+120)/(16c)$ to the scalar approximation $F_2^{\mathrm{scal}} = \kappa \cdot \lambda_2^{\mathrm{FP}}$. Whether the CohFT R-matrix produces compensating terms that restore the identity $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at genus $g \geq 2$ for multi-generator algebras remains open. At genus 1, the identity holds unconditionally for all families.

9.2. **The operadic complexity conjecture.** The four-class partition **G/L/C/M** classifies shadow depth, but its deeper origin is conjectural: the operadic complexity conjecture identifies $r_{\max}(\mathcal{A})$ with the A_∞-depth (the minimal n such that $m_k = 0$ for $k > n$ in the transferred A_∞ structure) and with the L_∞-formality level. The shadow-formality identification is proved at arities 2, 3, and 4; the full statement at all arities is open.

9.3. **Pixton's ideal and the shadow tower.** For class **M** algebras (infinite shadow tower), the Maurer–Cartan equation at each genus g produces a tautological relation on $\overline{\mathcal{M}}_g$. These relations lie in the Pixton ideal (the ideal of relations among tautological classes), and it is natural to ask whether the shadow tower of a single class **M** algebra generates the full ideal. Computational evidence at genus 2 and 3 supports this; a proof would require establishing ideal membership in the strata algebra.

9.4. **Arithmetic content of shadow denominators.** The Virasoro shadow coefficients S_r are rational functions of c with poles at $c = 0$ and $c = -22/5$. The denominators $c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$ grow in a pattern controlled by the Gaussian decomposition. For arities $r \geq 12$, the numerators may be divisible by Ramanujan-type primes whose appearance is governed by cusp forms. The total depth decomposition $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ separates the algebraic engine (depths ≤ 4) from the arithmetic frontier (depths ≥ 5).

9.5. **General DS primitivity.** Theorem 8.2 proves gravitational coproduct primitivity for the principal reduction $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$. The ghost-number obstruction argument uses only the universal grading of the BRST complex, and should extend to all principal reductions $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$ (Conjecture 8.3). Non-principal reductions require a separate analysis: the ghost complex has a more intricate grading, and the transferred coproduct may be nontrivial.

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